

ADAPTIVE DIGITAL CALIBRATION FRAMEWORK FOR MMWAVE SYSTEMS USING LEAST SQUARES ESTIMATION

ECC1394- SIGNAL PROCESSING FOR MM-WAVE COMMUNICATION

A PROJECT REPORT

Submitted by

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*in partial fulfilment for the award of the degree
of*

BACHELOR OF ENGINEERING in ELECTRONICS AND COMMUNICATION ENGINEERING



**K. RAMAKRISHNAN COLLEGE OF
ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
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BACHELOR OF ENGINEERING in ELECTRONICS AND COMMUNICATION ENGINEERING

**Under the Guidance of
Mrs. S. JANUPRIYA**

**Department of Electronics and Communication Engineering
K. RAMAKRISHNAN COLLEGE OF ENGINEERING**



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
Under
ANNA UNIVERSITY, CHENNAI**



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(An Autonomous Institution)

SAMAYAPURAM, TRICHY

BONAFIDE CERTIFICATE

Certified that this project report titled “**ADAPTIVE DIGITAL CALIBRATION FRAMEWORK FOR MMWAVE SYSTEMS USING LEAST SQUARES ESTIMATION**” is the bonafide work of **SHAHIN ASKIK K (8115U23EC097)**, **SHRUTHI R (8115U23EC102)** , **SRI AMIRTHA VARSHIBI S V (8115U23EC108)**, **SRI VIDHYA B (8115U23EC110)**, who carried out the project work under my supervision.

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DECLARATION BY THE CANDIDATE

We declare that to the best of our knowledge the work reported here in has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva - Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

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DEPARTMENT OF ECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will be competent enough to design, develop and integrate complex industrial products for modern electronics and communication systems.

PEO2: Graduates will be able to propose research oriented, industry relevant design solutions in state-of-the-art electronics and communication domains.

PEO3: Graduates will be able to engage in start-up and entrepreneurial activities developing innovative prototypes for the challenging needs of the society.

PROGRAM SPECIFIC OUTCOMES (PSO'S):

PSO1: Analyze and design complex electronic circuits and communication systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems,

industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, signal processing, high frequency and antenna design.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and the World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the

consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Lifelong learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

CO-PO MAPPING FOR PROJECT WORK

After successful completion of this course, the student should be able to

CO1: Analyze the Concept of Signal Models

CO2: Explain the principles of elements and functions involved in Spectrum Estimation

CO3: Apply the Optimum Filters concept in signal processing.

CO4: Develop the mm wave Communication Systems for 5G Communication system

CO5: Apply the mm-Wave MIMO in 5G and beyond

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	✓	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO3	✓	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO4	✓	–	✓	✓	✓	–	✓	✓	✓	✓	✓	✓	✓	✓
CO5	✓	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

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We thank **Dr. M. MAHESWARI M.Tech., Ph.D.**, Head of the Department of ELECTRONICS AND COMMUNICATION ENGINEERING, for providing her encouragement in pursuing this project.

We wish to convey our profound and heartfelt gratitude to our esteemed project guide **Mrs.S.JANUPRIYA M.Tech., (Ph.D.)**, Assistant Professor, Department of ELECTRONICS AND COMMUNICATION ENGINEERING, for her incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

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ABSTRACT

Millimeter wave (mmWave) communication has emerged as a key enabling technology for next-generation wireless systems such as 5G and future 6G networks due to its ability to provide extremely high data rates, large bandwidth, and low latency. Despite these advantages, the practical implementation of mmWave systems is significantly affected by hardware imperfections present in radio frequency (RF) components. These impairments include gain mismatch, phase offset, and in-phase/quadrature (I/Q) imbalance, which introduce distortion in the transmitted and received signals. As a result, system performance degrades in terms of increased bit error rate (BER), reduced signal quality, and improper constellation mapping. The method employs pilot-assisted transmission, where known reference signals are inserted into the data stream. A least squares (LS) estimation algorithm is then applied at the receiver to accurately estimate the impairment parameters such as amplitude imbalance and phase error. Once the impairments are estimated, appropriate compensation techniques are applied to correct the distorted signal and restore its original characteristics. The effectiveness of the proposed calibration method is evaluated through simulation using performance metrics such as BER and constellation diagrams. The results demonstrate a significant improvement in signal quality, with reduced distortion and enhanced symbol detection accuracy after calibration. Overall, this work highlights the importance of digital calibration techniques in improving the reliability and efficiency of mmWave communication systems. The proposed method provides a practical solution for mitigating hardware impairments and can be extended to advanced communication systems, including multi-antenna and adaptive wireless networks.

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CHAPTER 1

INTRODUCTION

Millimeter wave (mmWave) communication is an important technology used in 5G and future 6G networks due to its high data rate and large bandwidth. It operates at very high frequencies, enabling faster data transmission and low latency. However, practical mmWave systems face challenges due to hardware impairments in RF components. These impairments include gain mismatch, phase offset, and I/Q imbalance, which distort the signal. Such distortions lead to increased bit error rate and reduced system performance. Therefore, efficient calibration techniques are required to estimate and compensate for these impairments to improve communication quality.

1.1 Drawbacks of Existing System

- **High Hardware Complexity:** Existing systems depend on additional analog calibration circuits, which increase design complexity and make the system bulky.
- **High Cost:** Use of extra RF components and calibration hardware increases overall system cost.
- **Limited Accuracy:** Traditional methods cannot completely eliminate impairments such as I/Q imbalance and phase errors, leading to signal distortion.
- **Lack of Flexibility:** Hardware-based calibration cannot adapt easily to changing channel conditions and different operating environments.
- **No Real-Time Calibration:** Many existing systems do not support continuous or real-time adjustment of impairments during operation.

- **Increased Power Consumption:** Additional hardware components consume more power, reducing system efficiency.
- **Difficult Maintenance and Upgrades:** Hardware-based systems are harder to modify, maintain, and upgrade compared to software-based approaches.
- **Poor Performance at High Frequencies:** At mmWave frequencies, small hardware imperfections cause significant degradation, which existing systems fail to handle effectively.

1.2 Need for Proposed Work

Millimeter wave (mmWave) communication systems are a key technology in modern wireless networks such as 5G and future 6G due to their ability to support high data rates, large bandwidth, and low latency communication. However, practical implementation of mmWave systems faces significant challenges due to hardware imperfections present in RF components. These impairments, such as gain mismatch, phase offset, and I/Q imbalance, introduce distortion in the transmitted and received signals. This results in degraded signal quality, increased Bit Error Rate (BER), and reduced overall system performance.

- To reduce the impact of hardware impairments such as gain mismatch, phase errors, and I/Q imbalance in mmWave systems.
- To improve signal quality and system reliability by minimizing distortion and enhancing communication accuracy.
- To reduce Bit Error Rate (BER) and ensure efficient data transmission.
- To develop a cost-effective solution by replacing complex hardware calibration with software-based techniques.

- To enable real-time and adaptive calibration, allowing the system to adjust to varying channel conditions.

1.3 Applications of the Proposed System

- Used in 5G and future 6G wireless communication systems to improve signal quality and ensure high-speed data transmission.
- Applied in massive MIMO and beamforming systems, where accurate calibration is essential for proper signal alignment and transmission efficiency.
- Useful in satellite communication systems, where high-frequency signals require precise calibration to avoid distortion.
- Implemented in automotive radar and autonomous vehicle systems for accurate sensing and communication.
- Used in wireless backhaul networks to maintain reliable high-speed data links between base stations.
- Applicable in mmWave test and measurement systems for analyzing and improving communication performance.
- Can be integrated into IoT and smart communication networks for enhanced connectivity and reduced signal errors.
- Supports research and development in advanced wireless technologies, enabling innovation in next-generation communication systems.

CHAPTER -2

LITERATURE SURVEY

2.1 “Impact of Misalignments in mmWave MIMO Calibration Systems”

Author: C. Zhao et al.

Source: International Journal of Microwave and Wireless Technologies, 2025.

This paper studies calibration challenges in mmWave MIMO radar systems, focusing on errors caused by antenna misalignment. It shows that small misalignments significantly affect calibration accuracy and proposes improved calibration techniques for stable performance.

2.2 “Low-Complexity IQ Imbalance Compensation for mmWave Systems”

Author: Y. Zhang et al.

Source: IEEE / Research Publication, 2025.

This work presents a low-complexity digital architecture for compensating I/Q imbalance in communication systems. It reduces computational complexity while maintaining high accuracy in signal correction.

2.3 “Joint Estimation of IQ Imbalance and Power Amplifier Nonlinearity”

Author: X. Liu et al.

Source: Digital Signal Processing Journal, 2024.

This paper proposes an iterative estimation method to jointly compensate IQ imbalance and power amplifier distortion. The results show improved estimation accuracy and better system performance.

2.4 “IQ Imbalance Estimation for Optical and Wireless Systems”

Author: J. Zhou et al.

Source: Journal of Lightwave Technology, **2024**.

The study introduces advanced techniques for estimating IQ imbalance using digital signal processing methods, improving system flexibility and performance in high-speed communication networks.

2.5 “Carrier Phase Tracking with IQ Imbalance Compensation”

Author: S. A. Mir et al.

Source: Applied Optics, **2024**.

This paper combines phase tracking with IQ imbalance correction, demonstrating improved signal quality and reduced distortion in high-frequency communication systems.

2.6 “mmWave Channel Estimation Using Advanced Algorithms”

Author: J. P. González-Coma et al.

Source: IEEE Research, **2024**.

This paper focuses on channel estimation in mmWave MIMO systems using advanced algorithms, improving signal accuracy and system efficiency.

2.7 “IQ Imbalance Tolerant High-Speed Transmission System”

Author: S. Mir et al.

Source: Optical Communication Conference, **2023**.

The study demonstrates high-speed transmission systems that can tolerate IQ imbalance, improving communication reliability in high-frequency applications.

CHAPTER-3

PROJECT DESCRIPTION

This project focuses on millimeter wave (mmWave) communication system analysis for mitigating hardware impairments using a software-based calibration approach in MATLAB. The system estimates and compensates impairments such as gain mismatch, phase offset, and I/Q imbalance using pilot-based Least Squares (LS) estimation. It evaluates system performance through metrics like Bit Error Rate (BER) and constellation diagrams to ensure improved signal quality and reliability.

3.1 Work Flow and its Explanation

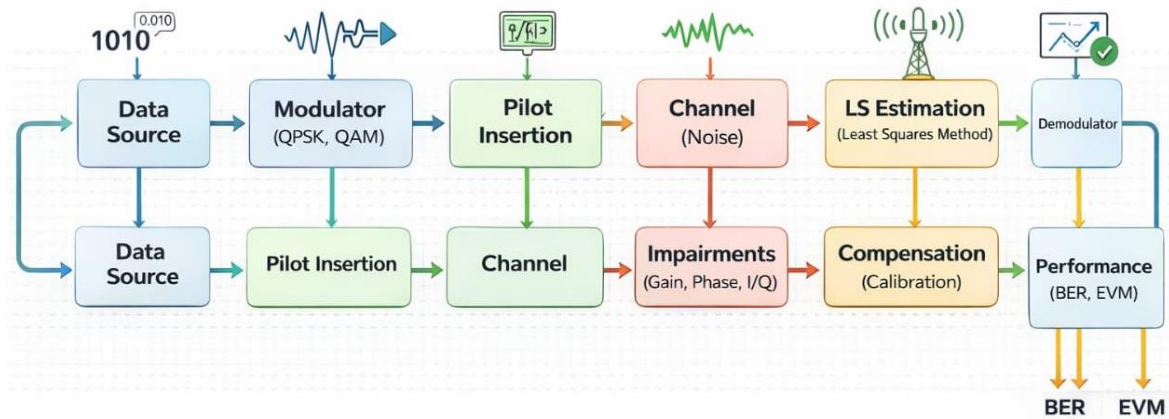


Fig. 3.1.1. Working flow diagram

3.2 Working Principle

The working principle of the proposed system is based on transmitting modulated data along with pilot signals through a mmWave channel affected by hardware impairments. At the receiver, these impairments are estimated using pilot-based Least Squares (LS) techniques. The estimated errors are then digitally compensated to improve signal quality and reduce bit error rate (BER).

3.2.1 Data Source

The data source is the first stage of the system where binary data is generated. It produces random bits (0s and 1s) that represent the information to be transmitted. These signals act as the input for further processing in the communication system.

3.2.2 Modulator (QPSK/QAM)

In this stage, the binary data is converted into modulated symbols. Techniques like QPSK or QAM are used to encode the data for efficient transmission. This helps in improving bandwidth utilization and transmission speed.

3.2.3 Pilot Insertion

Pilot signals are added to the modulated data at known intervals. These reference signals help in estimating channel conditions and impairments. They play an important role in accurate signal correction at the receiver.

3.2.4 Channel (Noise)

The signal passes through a communication channel during transmission. Noise and interference are introduced, affecting the signal quality. This simulates real-world wireless communication conditions.

3.2.5 Impairments (Gain, Phase, I/Q)

This stage introduces hardware imperfections into the signal. Impairments include gain mismatch, phase offset, and I/Q imbalance. These distortions degrade the signal and reduce system performance.

3.2.6 LS Estimation

The Least Squares method is used to estimate the impairments. It compares transmitted pilot signals with received signals to detect errors. This helps in identifying the exact distortion parameters.

3.2.7 Compensation (Calibration)

In this stage, the estimated errors are corrected digitally. The system compensates for gain, phase, and I/Q imbalance. This improves the signal quality and reduces distortion.

3.2.8 Demodulator

The corrected signal is converted back into binary data. It reverses the modulation process applied at the transmitter. This ensures recovery of the original transmitted information.

3.2.9 Performance (BER, EVM)

The final stage evaluates system performance. Metrics like Bit Error Rate (BER) and Error Vector Magnitude (EVM) are calculated. These values indicate the improvement achieved after calibration.

CHAPTER - 4

RESULTS AND DISCUSSION

4.1 Results

The output results of the proposed mmWave calibration system are presented by analyzing signal quality before and after compensation. Multiple output plots such as constellation diagrams and BER graphs are generated to demonstrate the consistency and effectiveness of the system. In the case of the uncalibrated signal, the constellation diagrams show scattered and distorted symbol patterns due to impairments like gain mismatch, phase offset, and I/Q imbalance. These distortions indicate poor signal quality and higher error rates. After applying the calibration process, the output results show well-aligned and clear constellation points. The symbols are tightly clustered, indicating that the impairments have been effectively reduced and the signal quality has improved significantly. Similarly, the Bit Error Rate (BER) results show a noticeable reduction after compensation.

4.2 Output Image

The output results of the proposed mmWave calibration system are analyzed to evaluate its performance in reducing signal distortions and improving communication quality. The system processes the transmitted signal, estimates impairments, and applies calibration using digital signal processing techniques. For each stage, output plots such as constellation diagrams, received signals, and BER graphs are presented. The comparison between uncalibrated and calibrated outputs clearly shows improvement in signal quality. The calibrated signal exhibits well-defined constellation points and reduced error rates, demonstrating the effectiveness of the proposed system in enhancing mmWave communication performance.

```

Enter number of symbols (e.g., 10000): 10110
Enter noise level (0.05 to 0.2): 0.3
Enter gain mismatch (e.g., 1.4): 1.3
Enter phase error in radians (e.g., pi/6): pi/2
Enter IQ imbalance (e.g., 0.25): 0.23

----- PERFORMANCE RESULTS -----
BER Before Calibration    = 0.9838
BER After Calibration     = 0.0454
EVM Before Calibration    = 1.7475|
EVM After Calibration     = 0.5327
>>

```

Fig. 4.2.1 Performance Results (BER and EVM Before and After Calibration)

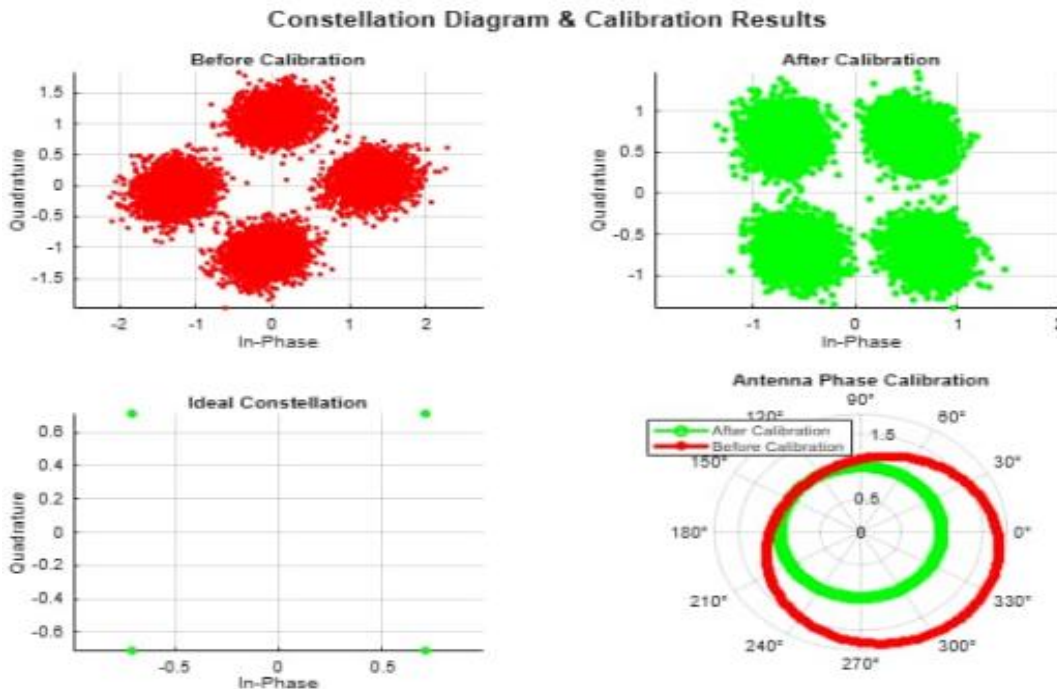


Fig. 4.2.2 Constellation diagram and Calibration Results

This figure shows the simulation input parameters and the performance results of the mmWave calibration system. It presents the values of Bit Error Rate (BER) and Error Vector Magnitude (EVM) before and after applying calibration. The results clearly indicate a significant reduction in both BER and EVM after calibration, demonstrating improved signal quality and system performance.

4.3 Discussion

The input parameters such as number of symbols (10110), noise level (0.3), gain mismatch (1.3), phase error ($\pi/2$), and I/Q imbalance (0.23) are not purely random but are intentionally chosen test values to simulate practical mmWave system conditions. These values represent moderate to high impairment scenarios, allowing the system performance to be clearly analyzed before and after calibration.

The Bit Error Rate (BER) is calculated by comparing the transmitted bits with the received bits after passing through the impaired channel. The formula used is **BER = Number of error bits / Total transmitted bits**. For example, if a large number of bits are affected due to noise and distortions, the BER becomes high (0.9838 before calibration). Similarly, the Error Vector Magnitude (EVM) is computed as the normalized difference between the ideal and received constellation points using **EVM = $\sqrt{(\sum|\text{error}|^2 / \sum|\text{ideal}|^2)}$** . The high EVM value (1.7475) indicates strong distortion in the received signal.

After applying pilot-based Least Squares (LS) estimation and compensation, the system corrects the impairments using calculated correction factors. Due to improved signal quality, the number of bit errors decreases significantly, resulting in a lower BER (0.0454), and the deviation from ideal constellation points reduces, giving a lower EVM (0.5327). Thus, the chosen parameter values help demonstrate the effectiveness of the proposed calibration method clearly.

CHAPTER – 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project demonstrates a software-based calibration approach for improving the performance of millimeter wave (mmWave) communication systems. The system effectively estimates and compensates hardware impairments such as gain mismatch, phase offset, and I/Q imbalance using pilot-based Least Squares (LS) estimation. The results show improved signal quality, with reduced Bit Error Rate (BER) and clearer constellation diagrams.

Overall, the proposed method is simple, cost-effective, and reliable compared to traditional hardware-based calibration techniques. It provides a strong foundation for enhancing communication performance and is suitable for modern wireless technologies such as 5G and future 6G systems.

5.2 Future scope

The proposed system can be further enhanced by implementing real-time calibration using hardware platforms such as FPGA or Software Defined Radio (SDR). This will improve practical applicability in real-world communication systems. Advanced estimation techniques, including machine learning and adaptive algorithms, can be incorporated to improve calibration accuracy under dynamic channel conditions.

The system can also be extended to multi-antenna (MIMO) and beamforming systems to support advanced 5G and 6G communication technologies. Additionally, integration with OFDM-based systems can further enhance performance in high-speed wireless networks. Future improvements may include real-time monitoring, automated adaptive calibration, and optimization for low-power communication devices.

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